

Stony Brook University
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Spring 2019

ASSIGNMENT 4: Buying and Selling, Labor Supply and Intertemporal consumption

Due: Monday April 15, in class.

INSTRUCTIONS: Show all your work or points will be deducted.

1. Suppose that there are 8,000 hours in a year and that an individual has a potential market wage of \$5 per hour. Initially, there is no non-labor income.
 - (a) What is the individual's full income?
 - (b) Suppose that preferences over relaxation and consumption are given by $u(c, R) = \sqrt{c} + \sqrt{R}$. How many hours will be worked?
 - (c) Suppose a rich uncle dies and leaves the individual an annual income of \$4000 per year. How many hours will be worked?
 - (d) How would your answer to part (c) change if the market wage were \$10 per hour instead of \$5 per hour?
 - (e) Graph's the individual's supply of labor curve implied by parts (c) and (d).
 - (f) Decompose the change in labor supply from part (d) after the wage change of $w = \$5$ to $w' = \$10$ into the substitution, income and endowment effects. Illustrate these effects in a graph.
2. Consider an agent that lives for three periods. In each period, the agents consumes c_t units of consumption for $t = 0, 1, 2$. Her utility for intertemporal consumption is given by $u(c_0, c_1, c_2) = \ln(c_0) + \delta \ln(c_1) + \delta^2 \ln(c_2)$, where $0 < \delta < 1$. Assume that this consumer has an initial wealth W that she receives in period 0 and that she receives no other income. A credit market allows her to save or borrow in the first two periods.

Let s_0 and s_1 and s_2 denote the amount that the consumer saves (or borrow if negative) in periods 0, 1 and 2 respectively.

 - (a) Write the per-period budget constraints for the consumer. Remember that the consumer receives wealth W in period 0.
 - (b) Provide an argument of why s_2 needs to be equal to 0 and nothing else in future periods.
 - (c) Using the fact that $s_2 = 0$ write the present value intertemporal budget constraint of this consumer by consolidating the per-period budget constraint as we saw in class.
 - (d) Solve for the intertemporal consumption choice of the agent, i.e. find $c_t^*(W, r, \delta)$ for $t = 0, 1, 2$. How do these demands depend on wealth W , the interest rate r and the parameter δ ? Can you provide an interpretation for δ ?
 - (e) Consider the following cases:
 - $(1 + r)\delta = 1$
 - $(1 + r)\delta < 1$
 - $(1 + r)\delta > 1$

What does each case imply about c_t^* ? Analyze each case by using your answer in part (d).